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From Scores to Steps:

Diagnosing and Improving LLM Performance in Evidence-Based Medical Calculations

Benlu Wang et al.

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Group K23

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Based on Wang et al., 2026, arXiv v2.

Presentation Outline

- 1 Motivation and Research Problem
- 2 Evaluation Framework
- 3 MedRaC Methodology
- 4 Paper Experiments and Results
- 5 Local Demo and Mini-Experiment
- 6 Discussion and Conclusion

From a plausible score to a verifiable calculation process.

Medical Calculations Require More Than Arithmetic

Clinical notes

Relevant and irrelevant details coexist.

Creatinine clearance

Formula + patient variables

Clinical question

Select the calculator that fits the question.

Corrected sodium

Formula + unit-aware arithmetic

Clinical consequence

An incorrect score can change a decision.

Wells score

Clinical criteria + score

Medical calculation combines interpretation, selection, extraction, and computation.

Final-Answer Accuracy Can Hide Invalid Reasoning

Corrected sodium: reference

$$Na_{\text{corrected}} = Na_{\text{measured}} + 0.024(\textit{Glucose} - 100) \quad (1)$$

Equation (1) gives 137.248 for $Na = 127$ and $\textit{Glucose} = 527$.

Incorrect model formula

$$127 + 0.016(527) = 135.432 \quad (2)$$

Final-answer evaluation

- ❶ Incorrect formula
- ❷ Numerically plausible answer
- ❸ Marked correct by broad tolerance

Formula: Incorrect

Final score accepted under the original tolerance.

A plausible answer is not necessarily a valid answer.

Research Questions and Contributions

Research questions

- ① How can each reasoning stage be evaluated?
- ② Where do LLM failures originate?
- ③ Can external tools reduce these errors?

Contributions

- ① Step-wise evaluation
- ② Structured error attribution
- ③ MedRaC: Formula RAG + Python execution

The paper evaluates, diagnoses, and then intervenes.

Medical Calculation as a Four-Stage Process

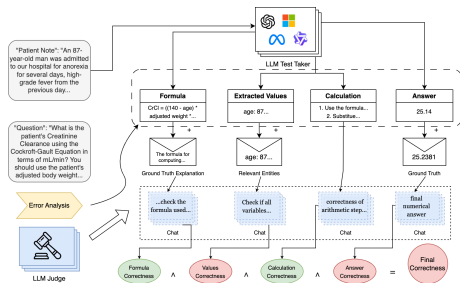


Figure 1: Four-stage evaluation of medical calculations.

F: Formula **E: Extraction** **C: Calculation** **A: Answer**
calculator method patient values valid operations reference result

Each stage can fail independently and propagate errors downstream.

Strict Correctness Requires Every Stage to Pass

Case-level criterion

$$\kappa = V(F) \wedge V(E) \wedge V(C) \wedge V(A) \quad (3)$$

$V(\cdot)$

Binary validation of a stage.

κ

Overall correctness for one case.

A correct case requires a valid Formula, Extraction, Calculation, and Answer.

Conditional Correctness Isolates Stage Reliability

Can stage S_i succeed after valid prerequisites?

$$CC_i = P(V(S_i) \mid V(S_1) \wedge \dots \wedge V(S_{i-1})) \quad (4)$$

S_i

The current stage in the F–E–C–A sequence.

Condition

All earlier stages must already be valid.

Conditional Correctness separates a stage's reliability from upstream failures.

First Error Attribution Identifies the Earliest Failure

Where does an incorrect case first break?

$$FE_i = P(V(S_1) \wedge \dots \wedge V(S_{i-1}) \wedge \neg V(S_i) \mid \neg \kappa) \quad (5)$$

$\neg V(S_i)$

Stage S_i is invalid.

$\neg \kappa$

The overall case is incorrect.

First Error Attribution directs improvement to the first failed capability.

Error Taxonomy Turns Failures into Diagnoses

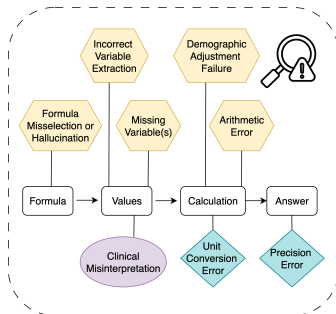


Figure 2: Error categories in medical calculations.

Knowledge

Formula selection; hallucination; demographic rules

Clinical interpretation

Incorrect extraction; missing variables; misinterpretation

Numerical execution

Unit conversion; arithmetic; rounding

Error attribution identifies which capability requires improvement.

MedRaC Grounds Formulas and Executes Calculations

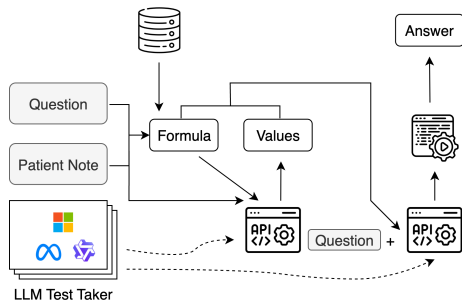


Figure 3: MedRaC architecture.

Retrieve formula
grounding

Extract values
clinical text

Generate code
formula to program

Execute code
deterministic result

The LLM interprets clinical text; tools ground formulas and compute deterministically.

Each MedRaC Component Targets a Distinct Failure Mode

Table 1: MedRaC components and targeted failures.

Component	Targeted failure
Formula RAG	Formula misselection and hallucination
Value extraction	Missing or incorrect inputs
Code generation	Formula-to-computation translation
Python execution	Arithmetic errors

Interpretation-dependent errors remain difficult.

Experimental Setup for Calculation and Clinical Scoring

Table 2: Paper experimental setup.

Benchmark	55 medical calculators; 1,048 original cases, 940 after cleaning
Task types	Equation-based and rule-based
Methods	Direct, CoT, One-shot, Self-Refine, MedPrompt, MedRaC
Models	Multiple open and closed LLMs
Evaluation	Final-answer and LLM step-wise evaluation

The benchmark tests both deterministic calculation and clinical scoring.

Main Results Across Models and Prompting Methods

Model	Direct		CoT		One-shot		Self-Refine		Medprompt		MedRaC	
	Rule	Calc	Rule	Calc	Rule	Calc	Rule	Calc	Rule	Calc	Rule	Calc
Phi-4-mini	16.52	2.16	5.01	2.16	12.09	16.47	2.06	6.16	3.24	10.32	35.10	68.39
LLaMA3.2-3B	12.39	0.83	1.47	3.99	14.75	18.47	0.88	2.83	2.95	16.31	⁻²	-
Qwen3-4B	23.30	26.79	9.73	25.79	49.85	59.57	9.44	26.29	10.32	45.92	45.72	68.72
Qwen3-8B	29.20	42.26	16.52	38.10	58.70	62.90	19.76	40.10	15.93	53.74	46.61	74.54
LLaMA3.1-8B	19.17	2.50	6.78	8.32	25.07	20.97	5.90	6.82	11.21	28.45	31.27	70.22
Qwen3-14B	39.53	46.59	26.55	43.43	60.77	67.05	27.14	47.25	8.55	22.30	50.44	78.37
GPT-4o-mini	22.42	7.99	23.89	34.61	52.80	49.58	26.55	33.94	11.80	42.43	50.44	72.71
GPT-4o	24.48	13.98	43.07	43.93	62.24	54.24	44.84	44.93	25.96	56.91	51.03	64.39

Table 3: Main performance results across models and methods.

Overall main result

MedRaC consistently improves calculation accuracy across the reported models, with the largest gains for smaller open models. Rule-based results are mixed, confirming that clinical interpretation remains the limiting capability.

Main Result: MedRaC Improves Equation-Based Tasks

Table 4: Paper-reported equation-based accuracy (%).

Model	One-shot	MedRaC	Gain
Phi-4-mini	16.47	68.39	+51.92
LLaMA3.1-8B	20.97	70.22	+49.25
GPT-4o	54.24	64.39	+10.15

Main finding

Formula grounding and deterministic execution increase equation-based accuracy by **10.15–51.92 percentage points**.

Rule-based gains remain mixed because clinical interpretation still dominates scoring.

Step-Wise Evaluation Aligns Closely with Expert Judgment

Table 5: LLM–Expert agreement across evaluation stages (%).

Agreement type	Formula	Extraction	Calculation	Answer
Expert–Expert	84.8	84.8	89.1	95.7
Expert–Non-Expert	72.3	78.1	66.6	91.3
LLM–Expert	90.2	78.3	88.1	97.8

Validation set

46 samples; five calculators.

Interpretation

Alignment is strongest for Formula and Answer; Extraction remains the hardest stage.

Error Reductions Reveal MedRaC's Remaining Boundary

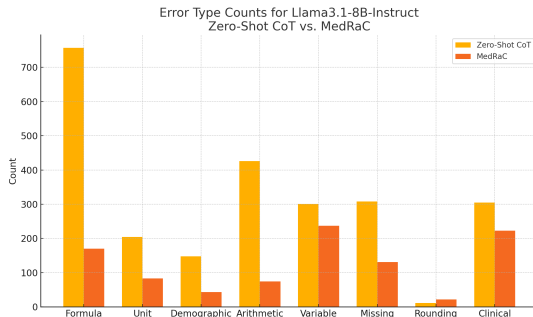


Figure 4: Error reduction with MedRaC.

Large reductions

Formula

Arithmetic

Demographic adjustment

Remaining boundary

Incorrect extraction

Clinical misinterpretation

MedRaC solves tool-addressable errors better than interpretation-dependent errors.

Local Live Demo Makes the F-E-C-A Pipeline Observable



Figure 5: Local live-demo pipeline.

Inspect

Intermediate outputs

Demonstrate

Complete pipeline

Compare

Plain CoT vs. MedRaC

The live run connects paper stages to observable outputs.

Local Mini-Experiment

Table 6: End-to-end rule evaluation.

Metric	Plain	MedRaC
Final answer	9/10	10/10
Completed	10/10	10/10

Table 7: Post-hoc LLM F–E–C–A evaluation.

Stage	Plain	MedRaC
Formula (F)	10/10	10/10
Extraction (E)	9/10	10/10
Calculation (C)	10/10	10/10
Answer (A)	8/10	9/10
Strict F–E–C–A	8/10	9/10

Fixed sample set: 10 cases (7 equation-based; 3 rule-based). Post-hoc evaluation uses the stored generation outputs.

Discussion

Limitations

- Structured, English-only cases
- Dependence on formula-bank quality
- Dependence on an LLM Judge
- Limited human validation

Future work

- Stronger clinical interpretation
- Broader human validation
- Richer formula knowledge
- Human-in-the-loop evaluation

Reliable medical AI requires tools and clinically grounded interpretation.

Conclusions

- ① Final-answer accuracy can mask clinically relevant reasoning errors.
- ② Step-wise evaluation makes Formula, Extraction, Calculation, and Answer failures measurable.
- ③ MedRaC improves formula grounding and arithmetic reliability; clinical interpretation remains open.

Reliable medical AI requires correct outputs and verifiable processes.

Thank you

Questions and Discussion

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